TATRASI AMAR

Software Engineer | Full-Stack Developer (Java · Spring Boot · React · Next.js)
+91 7330807277 | tatrasiamar123@gmail.com | Hyderabad, India | LinkedIn | GitHub | Portfolio

SUMMARY

Full-Stack Software Engineer with 2 years of experience at an early-stage product startup, building commercial multi-tenant B2B SaaS platforms (HRMS & Payroll) and real-time IoT architectures with Java, Spring Boot, React, TypeScript, Next.js, and PostgreSQL. Architected customizable payroll and biometric systems deployed for external client organizations, saving ₹3–4 Lakhs/yr in licensing. Experienced in Spring Security (JWT), Quartz scheduling, MQTT telemetry, and AI workflows. Immediate Joiner.

TECHNICAL SKILLS

- **Frontend:** React.js, Next.js, TypeScript, JavaScript, Tailwind CSS, Shadcn UI, Recharts, HTML5, CSS3
- **Languages & Backend:** Java, Spring Boot, Spring Security, Hibernate, RESTful APIs, Microservices, Python
- **Databases & ORM:** PostgreSQL, Supabase, MySQL, Drizzle ORM, Row-Level Security (RLS)
- **Real-Time & Distributed:** MQTT (Paho/HiveMQ), Quartz Scheduler, WebSockets, Modbus, SCADA/HMI
- **Cloud, AI & Tools:** Google Gemini API, Docker, Git, GitHub, Postman, CI/CD, Razorpay API, Agile/Scrum

PROFESSIONAL EXPERIENCE

Software Engineer (SE-1) | Terracos Technologies Pvt Ltd, Hyderabad | **Feb 2025 – Oct 2026**

- Engineered full-stack commercial B2B SaaS platforms (HRMS, Payroll, Industrial IoT BMS) with React, TypeScript, Spring Boot, and PostgreSQL, serving 1,000+ active enterprise users across client organizations.
- Architected a multi-tenant HRMS & payroll engine with dynamic salary slabs, tax formulas, and bank IFSC validation (100% calculation accuracy), commercially deployed for client companies.
- Implemented asynchronous MQTT telemetry ingestion for 10+ industrial sensors, cutting data latency by 50%.
- Integrated Quartz Scheduler with database persistence for automated nightly attendance aggregation and scheduled PDF payslip/email dispatches without blocking main threads.
- Built secure Razorpay payment flows with webhook handling for automated subscription billing, transaction verification, and invoice generation.
- Designed role-based access control (RBAC) dashboards with React, TypeScript, and Tailwind CSS; applied SSG optimizations, reducing page load times by 40%.
- Built Selenium automated testing workflows and collaborated in sprint planning and architectural code reviews (Agile/Scrum).

Software Engineer Intern | Terracos Technologies Pvt Ltd, Hyderabad | **Nov 2024 – Jan 2025**

- Developed responsive BMS dashboards for HVAC metrics, power distribution, and hardware alarms with Recharts and WebSockets.
- Contributed to PostgreSQL and Supabase schema design for the Attendance Management System, supporting real-time audit logs and tenant data isolation.
- Resolved frontend state management and API integration bottlenecks, improving dashboard rendering speed.

KEY PROJECTS

Terracos AI – Agentic AI B2B Platform *(Next.js 14, Google Gemini LLM, Drizzle ORM)*

- Built an autonomous B2B prospecting platform on Next.js 14 App Router and Gemini API that discovers, scores, and qualifies business leads.
- Implemented server actions, optimized PostgreSQL persistence with Drizzle ORM, and integrated transactional email via Resend API.

Enterprise HRMS & Payroll System *(Spring Boot, React, TypeScript, PostgreSQL)*

- Built a commercial multi-tenant B2B SaaS platform for biometric attendance, leave management, and automated payroll for 1,000+ employees.
- Implemented Controller-Service-Repository architecture, Spring Security JWT authentication, and Quartz cron jobs for scheduled reports.

BMS & Smart IoT Telemetry Platform *(Java, Spring Boot, MQTT, React, Recharts)*

- Built a live HVAC and power-meter telemetry pipeline with threshold-breach alerts, auto-reconnection, and historical data visualization.

EDUCATION

B.Tech, Computer Science & Engineering | Jayaprakash Narayan College of Engineering | **2021 – 2024**

Diploma, Electrical & Electronics Engineering | Jayaprakash Narayan College of Engineering | **2018 – 2021**

CERTIFICATIONS

Full Stack Development with Cloud Computing – EY-GDS | Data Structures and Algorithms in Java – GeeksforGeeks | Java Fundamentals – Oracle Academy | Cloud Computing – IBM | CCNA – Cisco | Claude 101 – Anthropic